Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: OJT Management System (Java Swing + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	OJT Management System
Owning group	[Owning group no. and name]
Repository	[Repository URL of the OJT Management System]
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt, a6_recovery.txt

Run on 6 November 2026 at 23:04 (Asia/Manila), after the application click-through of Activity 3. The original database and the dump were never modified.

2. Backup plan

Item	Plan
Objects	Whole database ojt_management: 12 tables, 12 sequences, constraints and grants
Method	Logical backup with pg_dump custom format (-Fc)
Account	The project defines no backup role, so the dump was taken as postgres. Recommend a role with pg_read_all_data
Schedule (real deployment)	Daily full dump plus a dump before each deployment
Retention	7 daily and 4 weekly copies on a second machine
RPO and RTO	RPO 24 hours, RTO 1 hour
Roles	Roles are cluster-wide and not in a database dump. Keep a roles-only dump (pg_dumpall --roles-only)
Sensitive content	The dump holds password hashes from the users table, so it must be stored with restricted access

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d ojt_management -f ojt_management_activity6.dump
```

Item	Value
File	ojt_management_activity6.dump
Size	54,021 bytes
SHA-256	F9149CE2218F1473CF70F872B0E40DAC945CB6961859D445919A9D8BF195CCE2

Archive	custom format, gzip, 143 TOC entries, 12 TABLE DATA entries, dumped by pg_dump 18.6
Errors	None, exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost ojt_management_restore
pg_restore -U postgres -h localhost -d ojt_management_restore ojt_management_activity6.dump
```

Restore succeeded: exit code 0 in 0.22 seconds, no warnings or errors.

Check	Original	Restored	Match
Tables, views, sequences	12, 0, 12	12, 0, 12	Yes
Row counts (users, students, academic_supervisors, company_supervisors, companies, requirements, student_requirements, ojt_applications, ojt_placements, attendance, evaluations, audit_log)	5, 1, 1, 1, 1, 3, 3, 1, 1, 1, 1, 0	same	Yes
Constraints: check, foreign key, not null, primary key, unique	10, 15, 80, 12, 12	same	Yes
Table grants	137	137	Yes
Triggers	0	0	Yes
Application connects to the restored database	not required	not tested	n/a

5. Access control after restore

Table and column grants came back with the dump. The database-level privileges did not: the restored database had an empty datacl until the CONNECT grant was applied to the five roles, after which it matched the original. As ojt_admin on the restored copy, SELECT on users returned 5 rows and SELECT on students was denied, the same as on the original.

6. Recovery simulation

Step	Action	Result
Before	Checksum of attendance in the restored copy	1 row, md5 d55742fab726b2abd6f4836d909b2a7d
Incident	DELETE of the only attendance row in the restored copy	DELETE 1, table now has 0 rows
Recovery	createdb ..._recovery and pg_restore of the same dump	Success, exit code 0
Compare	Checksum of attendance in the recovery database	1 row, same md5 d55742fa...
Original untouched	Checksum in the original database	1 row, same md5

Outcome: Recovered. A full restore into a fresh database was chosen because it leaves the damaged copy untouched for comparison. Data written after the last backup would be lost, which is the RPO.

7. Evaluation and recommendations

- Daily dumps are enough for a small training system. A system that records attendance and evaluations daily may want a shorter RPO, such as WAL archiving.
- The restore took 0.22 seconds, far inside the 1-hour RTO. Measure again as data grows.
- Keep a roles-only dump and the grant script with each backup and re-apply the database-level CONNECT privilege after each restore.
- Create a dedicated backup role and protect the dump files because they contain password hashes.

8. Limitations and errors

- Logical dumps capture one point in time only.
- The restore used the administrator account, so restored objects belong to postgres, the same as the original.
- No errors or warnings occurred. The only difference after restore was the missing database-level privileges.
- The application was not pointed at the restored database. The test databases stay until the evidence is captured and are then dropped.

9. Feedback sent to owners

Sent on 6 November 2026: the dump and restore worked without warnings and all objects and row counts matched. The database-level CONNECT privilege is not restored and must be re-applied. The project has no backup role.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own server, and every result in this report comes from the saved output files.